STEP 1 — Install Required Libraries

```
# Install required libraries if not already installed
!pip install spacy transformers torch pandas datasets

# Download English spaCy model
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
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Collecting en-core-web-sm==3.8.0
  WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) after connection broken by 'Prot
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-no
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✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

STEP 2 — Import Libraries

```
import spacy                    # NLP library for NER
from spacy import displacy      # To visualize named entities

from transformers import pipeline
from transformers import AutoTokenizer, AutoModelForTokenClassification

import torch                    # Deep learning framework
import pandas as pd             # For tables and DataFrames
```

STEP 3 — Load spaCy NER Model

```
# Load pretrained English spaCy model
nlp = spacy.load("en_core_web_sm")
```

STEP 4 — Prepare Sample Text Data

```
# List of real-world sentences
sentences = [
    "Apple launched the iPhone 15 in California on Tuesday.",
    "Virat Kohli scored a century against Australia in Chennai.",
    "The Prime Minister of India met the President of France in Paris.",
    "Google announced a new AI model at its New York event.",
    "Tesla plans to expand its manufacturing plant in Germany."
]
```

STEP 5 — Apply spaCy NER

```
# Loop through each sentence
for sentence in sentences:
    doc = nlp(sentence)          # Pass sentence to spaCy pipeline
    print("\nSentence:", sentence)

    # Extract named entities
    for ent in doc.ents:
        print("Entity:", ent.text, "| Label:", ent.label_)
```

```
Sentence: Apple launched the iPhone 15 in California on Tuesday.
Entity: Apple | Label: ORG
Entity: 15 | Label: CARDINAL
Entity: California | Label: GPE
Entity: Tuesday | Label: DATE

Sentence: Virat Kohli scored a century against Australia in Chennai.
Entity: Virat Kohli | Label: PERSON
Entity: a century | Label: DATE
Entity: Australia | Label: GPE
Entity: Chennai | Label: GPE

Sentence: The Prime Minister of India met the President of France in Paris.
Entity: India | Label: GPE
Entity: France | Label: GPE
Entity: Paris | Label: GPE

Sentence: Google announced a new AI model at its New York event.
Entity: Google | Label: ORG
Entity: AI | Label: GPE
Entity: New York | Label: GPE

Sentence: Tesla plans to expand its manufacturing plant in Germany.
Entity: Germany | Label: GPE
```

STEP 6 — Store spaCy Results in DataFrame

```
# Empty list to store results
spacy_results = []

# Extract entities and store them
for sentence in sentences:
    doc = nlp(sentence)
    for ent in doc.ents:
        spacy_results.append([sentence, ent.text, ent.label_])

# Create DataFrame
spacy_df = pd.DataFrame(
    spacy_results,
    columns=["Sentence", "Entity", "Label"]
)

# Display table
spacy_df
```

	Sentence	Entity	Label
0	Apple launched the iPhone 15 in California on ...	Apple	ORG
1	Apple launched the iPhone 15 in California on ...	15	CARDINAL
2	Apple launched the iPhone 15 in California on ...	California	GPE
3	Apple launched the iPhone 15 in California on ...	Tuesday	DATE
4	Virat Kohli scored a century against Australia...	Virat Kohli	PERSON
5	Virat Kohli scored a century against Australia...	a century	DATE
6	Virat Kohli scored a century against Australia...	Australia	GPE
7	Virat Kohli scored a century against Australia...	Chennai	GPE
8	The Prime Minister of India met the President ...	India	GPE
9	The Prime Minister of India met the President ...	France	GPE
10	The Prime Minister of India met the President ...	Paris	GPE
11	Google announced a new AI model at its New Yor...	Google	ORG
12	Google announced a new AI model at its New Yor...	AI	GPE
13	Google announced a new AI model at its New Yor...	New York	GPE
14	Tesla plans to expand its manufacturing plant ...	Germany	GPE

Next steps: Generate code with spacy_df New interactive sheet

STEP 7 — Visualize spaCy Entities (Optional)

```
# Visualize entities for first sentence
doc = nlp(sentences[0])
displacy.render(doc, style="ent", jupyter=True)
```

Apple ORG launched the iPhone 15 CARDINAL in California GPE on Tuesday DATE .

STEP 8 — Load Hugging Face Transformer NER Model

```
# Model fine-tuned for NER
model_name = "dbmdz/bert-large-cased-finetuned-conll03-english"

# Load tokenizer
tokenizer = AutoTokenizer.from_pretrained(model_name)

# Load model
model = AutoModelForTokenClassification.from_pretrained(model_name)

# Create NER pipeline
ner_pipeline = pipeline(
    "ner",
    model=model,
    tokenizer=tokenizer,
    aggregation_strategy="simple"   # Merges subword tokens
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

config.json: 100% 998/998 [00:00<00:00, 18.0kB/s]

tokenizer_config.json: 100% 60.0/60.0 [00:00<00:00, 1.21kB/s]

vocab.txt: 213k/? [00:00<00:00, 1.65MB/s]

model.safetensors: 100% 1.33G/1.33G [00:14<00:00, 168MB/s]

Loading weights: 100% 391/391 [00:00<00:00, 625.32it/s, Materializing param=classifier.weight]

```
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll03-english
Key                       | Status     |  |
-------------------------+------------+--+-
bert.pooler.dense.weight | UNEXPECTED |  |
bert.pooler.dense.bias   | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

STEP 9 — Apply Transformer NER

```
# Run NER on each sentence
for sentence in sentences:
    print("\nSentence:", sentence)
    results = ner_pipeline(sentence)

    for res in results:
        print(
            "Entity:", res["word"],
            "| Label:", res["entity_group"],
            "| Score:", round(res["score"], 2)
        )
```

```
Sentence: Apple launched the iPhone 15 in California on Tuesday.
Entity: Apple | Label: ORG | Score: 1.0
Entity: iPhone 15 | Label: MISC | Score: 1.0
Entity: California | Label: LOC | Score: 1.0

Sentence: Virat Kohli scored a century against Australia in Chennai.
Entity: Virat Kohli | Label: PER | Score: 1.0
Entity: Australia | Label: LOC | Score: 1.0
Entity: Chennai | Label: LOC | Score: 1.0

Sentence: The Prime Minister of India met the President of France in Paris.
Entity: India | Label: LOC | Score: 1.0
Entity: France | Label: LOC | Score: 1.0
Entity: Paris | Label: LOC | Score: 1.0

Sentence: Google announced a new AI model at its New York event.
Entity: Google | Label: ORG | Score: 1.0
Entity: New York | Label: LOC | Score: 1.0

Sentence: Tesla plans to expand its manufacturing plant in Germany.
Entity: Tesla | Label: ORG | Score: 1.0
Entity: Germany | Label: LOC | Score: 1.0
```

STEP 10 — Store Transformer Results in DataFrame

```
# Empty list to store transformer output
hf_results = []

# Extract entities and confidence scores
for sentence in sentences:
    results = ner_pipeline(sentence)
    for res in results:
        hf_results.append([
            sentence,
            res["word"],
            res["entity_group"],
            round(res["score"], 3)
        ])

# Create DataFrame
hf_df = pd.DataFrame(
    hf_results,
    columns=["Sentence", "Entity", "Label", "Confidence Score"]
```

```
)

# Display table
hf_df
```

	Sentence	Entity	Label	Confidence Score
0	Apple launched the iPhone 15 in California on ...	Apple	ORG	0.997
1	Apple launched the iPhone 15 in California on ...	iPhone 15	MISC	0.995
2	Apple launched the iPhone 15 in California on ...	California	LOC	1.000
3	Virat Kohli scored a century against Australia...	Virat Kohli	PER	0.996
4	Virat Kohli scored a century against Australia...	Australia	LOC	1.000
5	Virat Kohli scored a century against Australia...	Chennai	LOC	0.999
6	The Prime Minister of India met the President ...	India	LOC	0.999
7	The Prime Minister of India met the President ...	France	LOC	1.000
8	The Prime Minister of India met the President ...	Paris	LOC	1.000
9	Google announced a new AI model at its New Yor...	Google	ORG	0.999
10	Google announced a new AI model at its New Yor...	New York	LOC	0.999
11	Tesla plans to expand its manufacturing plant ...	Tesla	ORG	0.998
12	Tesla plans to expand its manufacturing plant ...	Germany	LOC	1.000

Next steps: Generate code with hf_df New interactive sheet

STEP 11 — Comparison Table

```
# Manual comparison table
comparison_df = pd.DataFrame({
    "Feature": [
        "Model Type",
        "Speed",
        "Accuracy",
        "Context Handling",
        "Confidence Score"
    ],
    "spaCy": [
        "Statistical + CNN",
        "Very Fast",
        "Good",
        "Limited",
        "No"
    ],
    "Hugging Face": [
        "Transformer (BERT)",
        "Slower",
        "Very High",
        "Excellent",
        "Yes"
    ]
})

comparison_df
```